BG02: Disturbance and tree responses

Reading Environment and History through Tree Rings | Lesson 6 supplement | English v1 | 30 October 2026

1. From leaves, roots and cambium to rings

Ring width measures a year's radial wood increment. Similar width changes can have different explanations. Connect organs and tissues with their functions before interpreting a ring pattern.

Part or tissue	Function	Question about disturbance
Leaf	Photosynthesis produces organic compounds.	When might foliage loss affect stem growth?
Root	Water and mineral uptake; anchorage.	What changes after root injury or burial?
Vascular cambium	Produces xylem and phloem.	How might growth record local injury?
Xylem	Water transport and support.	What can cell structure add beyond ring width?
Phloem	Transports sugars and other organic compounds.	Why identify the tissues attacked by each insect?

Responses are not a simple linear chain

Foliage loss and reduced stem growth need not occur immediately or proportionally. Consider stored resources, refoliation, growth phenology and competition. Neighbouring survivors may gain light or space after mortality. Damaged trees and their neighbours can respond differently.

Check the biological levels

Organs include leaves, stems and roots; tissues include xylem, phloem and cambium. Cell types include vessel elements and tracheids. Do not confuse these levels. Investigate features such as reaction wood in relation to posture changes and species.

Background: local Speer and Schweingruber books, Lesson 2 wood anatomy, and the shared BG01 supplement.

The plant body · Stems and cambium

2. Investigating insect damage with tree rings

Distinguish feeding modes

Aspect	What to check
Defoliation	Foliage loss, season, refoliation and recovery.
Sap feeding, boring and other damage	Species, attacked tissue, wounds, staining and pathogen associations.
Population effects	Host distribution, outbreak area, repetition and differences between survivors and dead trees.

Insects do not all attack the same tissues or leave the same ring pattern. Match identified agents and documented damage with ring observations where possible.

Three questions

1. Did host and non-host trees have comparable climate responses before the damage?

2. Do wood anatomy and damage records support the interpretation beyond narrow rings alone?

3. What is missed if only surviving trees are studied?

Evidence beyond width

Intra-annual density fluctuations (IADFs) are density or anatomical variations within a ring; they are not uniquely diagnostic of insects. Khasanov and Sandlersky (2018; R42) combined widths, anatomy and a non-host comparison.

In the hemlock study (R45), growth of survivors did not decline at every site. Revisit survivor bias, changing competition after neighbouring mortality, and possible physiological responses.

- R41 Cockchafer outbreaks and periodic narrow oak rings — Dendrochronologia (2013)
- R42 Does insect induced defoliation affect anatomical structure of oak wood? — Khasanov & Sandlersky (2018)
- R45 Radial growth changes following hemlock woolly adelgid infestation — Walker et al. (2014)

3. Eruptions, earthquakes and tree responses

An eruption example

At Mount St. Helens in 1980, blast, tephra and deposits produced spatially different damage and survival. Dendrovolcanology examines suppression, scars, mortality and establishment together with deposits and terrain, connecting to dendrogeomorphology.

An independently dated event such as 1980 helps investigate tree responses. Applying those observations to earlier eruption dating requires attention to preservation and alternative causes. Distinguish direct local impacts from broader climatic effects of volcanism.

An earthquake example

Dendroseismology examines tree responses to damaged roots or crowns, tilting and changed water conditions. At Wrightwood, growth suppression from 1813 was associated with the late-1812 earthquake.

Date or interval	Interpretation
Earthquake timing	Assessed using historical records, sediments and other evidence.
First tree response	Can occur in the following year because of season or lag.
Non-recording interval	Does not demonstrate the absence of earthquakes.

Compare near-fault and off-fault trees, terrain and other disturbances. Ring changes alone do not establish every earthquake, its size or its recurrence timing.

- R46 Mount St. Helens: a living laboratory — US Forest Service
- R47 Irregular recurrence of large earthquakes: evidence from trees — Jacoby et al. (1988)
- R48 The 1812 Wrightwood earthquake: tree rings and other evidence — Caltech SCEDC

Japanese-language review: Hoshino and Oikawa (2011), dendrochronological dating of volcanic activity. Compare living and dead trees, direct disturbance and climate responses.

4. A reading sheet for images, papers and data

Separate observation and inference

Entry	What to write
Observation	Direct features: widths, cells, scars, staining and location.
Hypothesis	Candidate explanations such as insects, climate, fire, root damage or competition.
Independent evidence	Damage records, weather, photographs, sediments, earthquake or eruption records, comparison trees.
Limitations	Missing rings, non-recording, sampling bias, comparison differences and unresolved information.

Choose an image from The Nature Conservancy photo essay (R50) or the USGS specimen account (R43). Complete these four entries. Separate visible features from explanations supplied by captions or research. View the photographs in the original article.

Check a graph

- Does the vertical axis show width, index, area, counts or percentages?
- What is the denominator? Not recording is not zero.
- Do two 100% increases have the same baseline and absolute change?
- Does repetition alone establish a calendar date or cause?

Next step

Choose a hypothesis, state an observation consistent with it, and explain why an alternative remains possible. Propose one additional observation that could change the conclusion. Use the result in the course research plan or review.

- R43 A whitebark pine cross section: fire, insects and death — USGS (2023)
- R50 Reading the Tree Rings: a photo essay — The Nature Conservancy (2026)